

Decentralized Public R&D, Demand Shifters, and the Over- and Underinvestment Threshold

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Abstract

We study decentralized public R&D by local governments when downstream firms compete in prices and policy shifts demand. In a two-jurisdiction Hotelling model, public R&D raises perceived quality that combines a welfare-relevant (real) component and a purely persuasive component. The welfare-relevant share α determines how much of perceived quality is counted in welfare, while spillovers are governed by ρ . Solving downstream Bertrand pricing yields a reduced-form profit function, enabling a closed-form comparison between the decentralized Nash equilibrium and the social optimum. We obtain a sharp sign reversal: decentralized investment is excessive if and only if α is below the cutoff $\bar{\alpha}(\rho) = 2(1 - \rho)/[3(1 + \rho)]$, and is insufficient above it; in the purely persuasive benchmark ($\alpha = 0$), the first best entails zero investment. We also show that the social optimum can be implemented by a single matching instrument on R&D costs (allowing for negative matching), with an implementing rate $s^* = 1 - \bar{\alpha}(\rho)/\alpha$ in closed form. The results clarify when interjurisdictional competition becomes a promotion-policy trap and when knowledge spillovers instead call for coordinated support.

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1 Introduction

Across the world, subnational governments spend public money to “upgrade” and market local products. Consider a concrete example that is easy to visualize: two neighboring coffee-growing regions that each want to position their beans as *specialty* in export

markets. Both regional governments fund agronomic and processing programs (e.g., disease-resistant varieties, fermentation protocols, quality-control labs, and certification/traceability systems) and also fund branding campaigns (logos, events, slogans, influencer marketing, and tourism tie-ins).¹ These interventions raise consumers’ *perceived* quality of a region’s coffee and can be leveraged by downstream firms (e.g., cooperatives and trading firms) competing in prices. When many jurisdictions pursue similar programs, policy becomes a strategic instrument in *interjurisdictional competition* for demand and local rents. The resulting externalities—business stealing on one side, knowledge spillovers on the other—share a family resemblance with the fiscal externalities in the tax-competition literature (Tiebout 1956; Wilson 1986; Zodrow and Mieszkowski 1986; Wildasin 1988; Keen and Marchand 1997), though the mechanism here operates through *product-market demand shifting* rather than competition for mobile capital.² A practical policy question follows: when does decentralized public “R&D-plus-promotion” create a *promotion-policy trap*—costly spending that mostly reallocates demand across regions—and when does it instead lead to underinvestment because knowledge spillovers are not internalized?³

A central difficulty is welfare evaluation when policy shifts demand. Some policy-induced demand shifts reflect real improvements in product quality (better taste, safety, reliability, or verifiable attributes), while others reflect persuasion that changes perceptions without improving underlying quality. We capture this distinction parsimoniously by decomposing perceived quality into a welfare-relevant (real) component and a purely persuasive component. A single parameter $\alpha \in [0, 1]$ indexes the *welfare-relevant share*: how much of perceived quality is counted in welfare. This connects our setting to classic welfare analyses of advertising and demand-shifting activities (Dorfman and Steiner 1954; Dixit and Norman 1978) and to the informative-versus-persuasive distinction in advertising economics (Nelson 1974; Grossman and Shapiro 1984; Becker and Murphy 1993; Akerberg 2001; Bloch and Manceau 1999). Intuitively, when α is low (spending is mostly branding/persuasion), interjurisdictional competition tends to look like business stealing; when α is high (spending is mostly quality improvement and verification), the same competition can be socially valuable, especially if knowledge spills over.

We study this trade-off in a two-jurisdiction Hotelling model in which downstream firms compete in prices (Bertrand) and policy affects demand through perceived quality. Public investment creates a perceived-quality index that spills over across jurisdictions at rate $\rho \in [0, 1]$. Solving the downstream pricing game yields a reduced-form profit function, which allows a closed-form characterization of both the decentralized Nash equilibrium and the social optimum.

¹In many agricultural settings, research and promotional activities are administered by *separate* institutions—for example, public experiment stations versus commodity-promotion boards (checkoffs). Our model abstracts from this institutional separation by treating the jurisdiction as a single decision-maker. The two-input extension in Section 5.6 introduces separate decision variables for research and promotion, which partially addresses this concern.

²In the standard tax-competition framework, jurisdictions attract mobile capital by lowering tax rates, generating fiscal externalities. Here, no factor is mobile; instead, public investment shifts demand among differentiated products. The paper thus sits at the intersection of R&D competition models (d’Aspremont and Jacquemin 1988; Kamien et al. 1992) and intergovernmental-grant design (Wildasin 1984, 1989), with public rather than private actors choosing the demand-shifting investment.

³Many real-world programs bundle welfare-relevant research with generic promotion; for beggar-thy-neighbor incentives in promotion programs, see Alston et al. 2001.

Our main results can be stated succinctly:

- **Closed-form sign reversal (threshold).** Decentralized public R&D is *excessive* if and only if the welfare-relevant share α is below a cutoff that depends on spillovers,

$$\bar{\alpha}(\rho) \equiv \frac{2(1 - \rho)}{3(1 + \rho)}. \quad (1)$$

It is *insufficient* if $\alpha > \bar{\alpha}(\rho)$. Figure 1 summarizes the regions of over- and under-investment and can be read as a policy-diagnostic chart: the “overinvestment” side corresponds to persuasion-heavy (branding/promotion) packages, while the “underinvestment” side corresponds to research-heavy (real-quality) packages with spillovers.

- **Single-instrument implementation benchmark.** Within the one-dimensional baseline model, a single matching instrument on R&D costs decentralizes the constrained first best with an implementing rate $s^* = 1 - \bar{\alpha}(\rho)/\alpha$ in closed form. The required match can be negative, reflecting that when α is sufficiently small, the benchmark calls for restraining (rather than subsidizing) decentralized spending.
- **Microfoundation for α (two-input extension).** We provide a parsimonious extension in which policy has two components: welfare-relevant research and welfare-irrelevant promotion. The extension delivers closed-form allocations and implies that decentralization tilts spending toward persuasion (a *promotion tilt*), thereby lowering the implied welfare-relevant share in closed form. This both clarifies the interpretation of the promotion-policy trap and addresses the concern that α is merely a reduced-form welfare weight.

The threshold has a simple economic logic. A smaller welfare-relevant share magnifies redistributive incentives: when policy mainly persuades rather than improves quality, each jurisdiction has a strong motive to invest in order to steal demand from its rival. By contrast, stronger spillovers raise the social value of investment relative to private local returns, pushing the distortion toward underinvestment. In the coffee example, investments in agronomy and verification (quality-control labs, traceability, safety standards) are plausibly high- α , whereas spending on slogans, events, and image marketing is plausibly low- α .⁴ The model predicts that these two program types can lie on opposite sides of Figure 1: the same interjurisdictional rivalry may be socially productive in one case and socially wasteful in the other.

Finally, although our analysis is theoretical, the key objects admit an empirical interpretation that makes the cutoff operational in principle. The welfare-relevant share α can be proxied by how much a policy package translates into *objective* quality improvements versus *perception-only* shifts. High- α components include investments that measurably improve or verify quality (e.g., agronomic and process R&D, safety and quality-control infrastructure, certification and traceability systems), while low- α

⁴Geographical indications (GIs) and related certification schemes (PDO/PGI in the EU, GI marks in East Asia) can be interpreted as institutional devices that *raise* the effective α by tying the brand to verifiable production standards. By restricting promotional claims to those backed by certified attributes, GIs also function as *spillover-governance* mechanisms, partly controlling ρ through rules on permitted production area and methods (Crespi and Marette 2003). The interplay between GI institutions and the incentives analyzed here is an interesting avenue for future work.

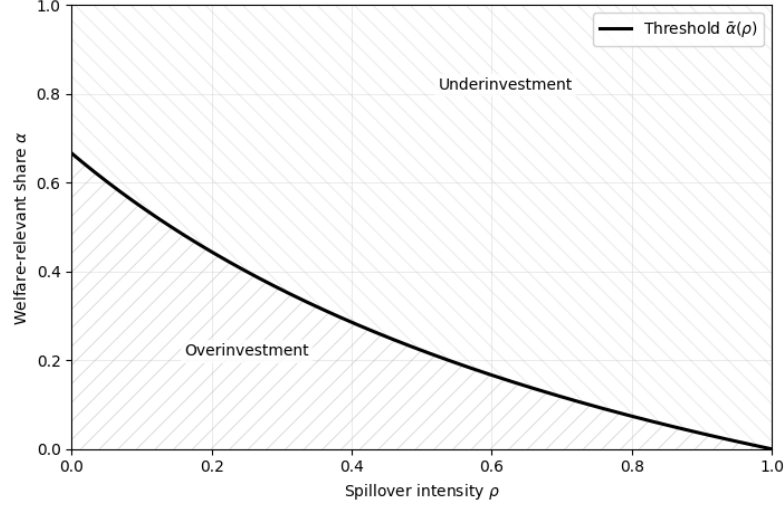


Fig. 1 Regions of over- and underinvestment as a function of spillovers (ρ) and the welfare-relevant share (α). The solid curve plots the closed-form threshold $\bar{\alpha}(\rho) = 2(1 - \rho)/[3(1 + \rho)]$ derived in Proposition 4.

components include spending whose primary channel is persuasion (e.g., branding campaigns, slogans, promotional events, influencer marketing). Empirically, α is thus related to (i) expenditure shares on verifiable quality-improvement/verification versus promotion, and (ii) outcomes such as objective quality scores, certification uptake, safety incidents, and price premia for verified attributes, contrasted with survey-based perception shifts unaccompanied by objective changes. Spillovers ρ can be proxied by cross-region diffusion of practices and know-how (e.g., adoption patterns, shared extension and training networks, or other diffusion indicators). These mappings motivate our focus on a closed-form cutoff and a single-instrument rule: the key policy risk is misclassifying the environment (subsidize versus restrain), which is most salient when α is low and spillovers are moderate.

Relative to canonical spillover-driven R&D models (Arrow 1962; d’Aspremont and Jacquemin 1988; Katz 1986; Kamien et al. 1992; Leahy and Neary 1997; Spence 1984) and persuasive-advertising models in differentiated markets (Bloch and Manceau 1999; Bagwell 2007), the novelty is to combine knowledge spillovers and business-stealing incentives *within a decentralized policy problem* and to close the loop from (i) equilibrium distortion to (ii) implementable intergovernmental matching design (Wildasin 1984, 1989). In addition, Section 5 provides a compact two-input extension that microfound α by decomposing policy into welfare-relevant research and welfare-irrelevant promotion while preserving the reduced-form profit dependence on the perceived-quality gap.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 derives the downstream price equilibrium and the reduced-form profit function. Section 4 characterizes the decentralized R&D equilibrium. Section 5 compares the decentralized outcome to the social optimum, derives the threshold result and

implementation, and then presents a two-input extension that microfound α via real R&D and promotion. Section 6 concludes.

2 Model

There are two jurisdictions (local governments) indexed by $i \in \{1, 2\}$, and we write $j \neq i$ for the other jurisdiction. Consumers are uniformly distributed on the Hotelling line segment $x \in [0, 1]$, and each consumer demands one unit. Throughout, we assume $\alpha \in [0, 1]$ and $\rho \in [0, 1]$.

2.1 Downstream actors (downstream firms)

To avoid confusion with jurisdiction indices, we label the two downstream sellers (downstream firms) by $f \in \{A, B\}$ and write their prices as (p_A, p_B) . Jurisdiction 1 is associated with seller A , and jurisdiction 2 with seller B . Marginal cost is constant and equal to c (upstream farmers are competitive, so the supply side is summarized by c).

2.2 Perceived quality and consumer utility (decomposing real vs. purely persuasive components)

Jurisdictional public R&D (cultivar breeding, branding, promotion, etc.) generates a quality component that shifts consumers' *perceived utility*. We refer to this demand shifter as perceived quality (q_A, q_B) .

We interpret q_f as *perceived quality*. A fraction $\alpha \in [0, 1]$ is *real quality* that enters welfare, $r_f \equiv \alpha q_f$, while the remainder⁵ $q_f - r_f = (1 - \alpha)q_f$ is a *purely persuasive component* that affects demand but not welfare. We refer to α as the *real-quality share*.⁶

When consumer x purchases good $f \in \{A, B\}$, her (private) utility is

$$U_f(x) = V + \bar{q} + q_f - p_f - t d_f(x), \quad (2)$$

where $t > 0$, $d_A(x) = x$, and $d_B(x) = 1 - x$. We assume V is sufficiently large so that the market is fully covered (every consumer purchases from one of the two sellers).

This decomposition implies that $\alpha = 0$ corresponds to a purely persuasive component (pure demand reallocation), whereas $\alpha = 1$ corresponds to purely real quality (a genuine welfare improvement).

⁵While α is introduced here as a reduced-form welfare-relevant share of perceived quality, Section 5 provides a simple two-input extension that microfound α by decomposing policy into welfare-relevant research and welfare-irrelevant promotion. In that extension, decentralization generates a promotion tilt and lowers the implied welfare-relevant share in closed form, addressing concerns that α is an ad hoc welfare weight.

⁶An alternative welfare criterion, due to Becker and Murphy (1993), treats advertising as a complementary good that enters consumers' utility directly and is therefore welfare-relevant by construction. Under this view, any demand shift—including pure persuasion—would be welfare-improving, effectively setting $\alpha = 1$ for all components. Our approach follows the Dixit–Norman tradition (Dixit and Norman 1978) in which welfare counts only *real* quality improvements. When both perspectives are plausible, one would replace our single α with a weight that interpolates between the two criteria; the qualitative over-/underinvestment threshold would persist but at a different cutoff.

$$r_f \equiv \alpha q_f, \quad q_f - r_f = (1 - \alpha)q_f. \quad (3)$$

(1) Important remark: Second-stage price competition, demand, and profits are determined by the overall demand shifter q_f that enters consumers' utility. The decomposition in (3) is introduced *solely* to evaluate welfare by counting real quality r_f but excluding the purely persuasive component $q_f - r_f$. Accordingly, the second-stage price equilibrium is expressed in terms of q_f (and the difference Δq). We treat α as reduced-form in the baseline; Section 5.6 provides a microfoundation by decomposing policy into real and purely persuasive components while preserving the same reduced-form profit dependence on Δq .

2.3 R&D and the production of perceived quality

Jurisdiction $i \in \{1, 2\}$ chooses an R&D investment level $e_i \geq 0$, which generates perceived quality for the associated seller via

$$q_A = g(e_1) + \rho g(e_2), \quad q_B = g(e_2) + \rho g(e_1), \quad (4)$$

where $g'(e) > 0$ and $g''(e) \leq 0$ (diminishing marginal returns), and where $\rho \in [0, 1]$ captures knowledge spillovers.⁷ The R&D cost is $K(e_i)$, with $K(0) = 0$, $K'(e) > 0$, and $K''(e) > 0$. Finally, for budgetary or administrative reasons, we restrict $e_i \in [0, \bar{e}]$ for some $\bar{e} > 0$ (the upper bound is nonbinding if $\bar{e}$ is sufficiently large).

2.4 Timing and objective functions (a central-government coordination mechanism)

The timing is as follows. (0) A central government (coordination mechanism) chooses a matching rate s (allowing $s < 0$ captures a tax). (1) Jurisdictions choose (e_1, e_2) . (2) Downstream actors (sellers A and B) simultaneously choose (p_A, p_B) . (3) Consumers choose where to buy.

Jurisdiction $i \in \{1, 2\}$'s decentralized objective is to maximize local downstream profit net of the (post-matching) effective R&D cost,

$$W_i^{\text{loc}} = \pi_{f(i)}(p_A, p_B; q_A, q_B) - (1 - s)K(e_i), \quad f(1) = A, \quad f(2) = B, \quad (5)$$

with respect to e_i .

(2) Admissible set for the matching rate: We allow $s < 0$ (a tax), but we assume at least that the jurisdiction's effective cost share is nonnegative, namely

$$s \leq 1 \quad (\text{equivalently, } 1 - s \geq 0). \quad (6)$$

⁷We restrict $\rho \geq 0$, treating spillovers as nonnegative knowledge transfers. In practice, a rival's quality improvement may *hurt* a jurisdiction through demand reallocation, which is captured here through the competitive interaction (the Δq term in profits) rather than through the sign of ρ . Three asymmetry considerations are worth noting: (i) research spillovers are typically nonnegative and depend on adaptation costs (Alston et al. 1995); (ii) promotional spillovers may be negative if a rival's campaign crowds out attention, or positive if it expands category awareness; and (iii) the symmetric ρ assumption abstracts from geography, absorptive capacity, and institutional heterogeneity. Allowing distinct spillover parameters $\rho_R \neq \rho_M$ is straightforward within our two-input extension (Section 5.6).

Accordingly, the admissible set is $s \in (-\infty, 1]$. We rule out $s > 1$ because it would imply a negative effective cost for the jurisdiction.

Financing assumption: We assume that the government’s budget impact of the subsidy (or tax) is offset through lump-sum taxes/transfers. Hence, s is a *pure transfer instrument* that changes marginal incentives without generating additional real resource costs or deadweight losses.

2.5 Notation (summary)

We use $x \in [0, 1]$ for consumer location, e_i for jurisdiction i ’s R&D effort, q_f for perceived quality, and $r_f \equiv \alpha q_f$ for real quality. The spillover parameter is ρ , and s denotes the matching rate (possibly negative, i.e., a tax).

Social welfare is evaluated by excluding the purely persuasive component $q_f - r_f$ and counting only real quality r_f , as in (3).⁸ Specifically, dropping the constant term $V + \bar{q}$, define

$$W^{\text{soc}} = \int_0^1 r_{\sigma(x)} dx - \int_0^1 t d_{\sigma(x)}(x) dx - c - K(e_1) - K(e_2), \quad (7)$$

where $\sigma(x) \in \{A, B\}$ denotes the good purchased by consumer x (equivalently, the seller chosen) in the downstream equilibrium.

(4) Fixing the welfare criterion: In (7), we count only real quality $r_f = \alpha q_f$ and exclude the purely persuasive component $q_f - r_f = (1 - \alpha)q_f$; that is, perceived quality affects welfare only through its real part.

(5) Consumer surplus and the welfare function: Under full market coverage (every consumer buys from one seller), consumer surplus is

$$CS = \int_0^1 [V + \bar{q} + q_{\sigma(x)} - p_{\sigma(x)} - t d_{\sigma(x)}(x)] dx.$$

Price payments are pure transfers between consumers and firms and therefore cancel in a welfare aggregate that includes producer profit.⁹ Consequently, the welfare function (7) captures the net social value: real quality received by consumers minus the aggregate transport cost and R&D expenditure.

⁸Because the market is fully covered ($D_A + D_B = 1$), the model excludes a *market-expansion* effect of promotion. In many agricultural contexts, generic promotion aims to expand total category demand (e.g., increasing per-capita milk or beef consumption). Introducing a variable-coverage specification would add a welfare-relevant market-creation channel even for persuasion, thereby reducing the overinvestment region. This is left for future work.

⁹If the planner also values consumer surplus explicitly—for example, under a more general objective $W = (1 - \theta) \cdot CS + \theta \cdot \sum_i (\pi_i - K_i)$ with $\theta \in (0, 1)$ —the threshold $\bar{\alpha}(\rho)$ would be modified but the qualitative over-/underinvestment logic would remain. We adopt the profit-based jurisdictional objective because it captures the incentive of local governments that act as revenue maximizers for their regional industries, as is common in the agricultural-promotion literature (Alston et al. 2001).

3 Price Competition

Lemma 1 (Indifferent consumer and demands). *Given (p_A, p_B) and (q_A, q_B) , the indifferent consumer x^* is*

$$x^*(p_A, p_B; q_A, q_B) = \frac{q_A - q_B - p_A + p_B + t}{2t}. \quad (8)$$

If $x^* \in (0, 1)$, demands are $D_A = x^*$ and $D_B = 1 - x^*$.

Lemma 2 (Interiority around symmetry and legitimacy of the reduced form). *Assume full market coverage (every consumer purchases from one of the two sellers). Let the perceived-quality differential be $\Delta q(e_1, e_2) \equiv q_A - q_B$, where (q_A, q_B) are generated from (e_1, e_2) by (4), and assume that g is continuous.*

(1) *At any symmetric profile $e_1 = e_2$, we have $\Delta q = 0$ and the second-stage price equilibrium is interior. In particular, $p_A^* = p_B^* = c + t$ and $D_A^* = D_B^* = 1/2$.*

(2) *For any symmetric point $(\bar{e}, \bar{e})$, there exists $\varepsilon > 0$ such that, for all (e_1, e_2) satisfying $\|(e_1, e_2) - (\bar{e}, \bar{e})\| < \varepsilon$, we have $|\Delta q(e_1, e_2)| < 3t$. Hence, in this neighborhood, the interior price equilibrium exists uniquely, and the reduced-form profit expressions (Corollary 1) as well as differentiation via the chain rule are valid.*

(3) **(A self-contained sufficient condition.)** *If $e_i \in [0, \bar{e}]$, then*

$$\Delta q(e_1, e_2) = (1 - \rho)\{g(e_1) - g(e_2)\}$$

implies

$$\max_{e_1, e_2 \in [0, \bar{e}]} |\Delta q(e_1, e_2)| \leq (1 - \rho)\{g(\bar{e}) - g(0)\}.$$

Therefore, if

$$(1 - \rho)\{g(\bar{e}) - g(0)\} < 3t, \quad (9)$$

then $|\Delta q| < 3t$ holds over the entire feasible set, and Proposition 1 and Corollary 1 apply globally (not merely locally).

Intuition. The interiority restriction $|\Delta q| < 3t$ ensures that the indifferent consumer lies strictly inside the Hotelling segment, so demands are well defined and the standard first-order conditions characterize a unique interior price equilibrium.¹⁰

Proof See Appendix A (Proof of Lemma 2). □

Proposition 1 (Price equilibrium given perceived qualities). *On the interior region guaranteed by Lemma 2 (in particular, $|\Delta q| < 3t$), the downstream price game admits*

¹⁰When $|\Delta q| \geq 3t$, one seller captures the entire market (a corner solution). This can occur if t is sufficiently small relative to quality differences—that is, if consumers care much more about quality than about “location” (brand loyalty). In such a regime, the price equilibrium takes a limit-pricing form and the smooth comparative statics of our analysis break down. We maintain the interiority assumption throughout; Lemma 2 provides an explicit sufficient condition under which it holds globally.

a unique Nash equilibrium,

$$p_A^*(q_A, q_B) = c + t + \frac{q_A - q_B}{3}, \quad (10)$$

$$p_B^*(q_A, q_B) = c + t - \frac{q_A - q_B}{3}. \quad (11)$$

Equilibrium market shares are

$$D_A^* = \frac{1}{2} + \frac{q_A - q_B}{6t}, \quad D_B^* = \frac{1}{2} - \frac{q_A - q_B}{6t}. \quad (12)$$

Proof See Appendix A (Proof of Proposition 1). $\square$

Corollary 1 (Equilibrium profits (reduced form)). *On the interior region in Lemma 2, substituting the equilibrium in Proposition 1 into profits yields*

$$\pi_A^*(q_A, q_B) = \frac{(3t + q_A - q_B)^2}{18t}, \quad \pi_B^*(q_A, q_B) = \frac{(3t - q_A + q_B)^2}{18t}. \quad (13)$$

Proof Let $\Delta q \equiv q_A - q_B$. By Proposition 1,

$$p_A^* - c = t + \frac{\Delta q}{3}, \quad D_A^* = \frac{1}{2} + \frac{\Delta q}{6t}.$$

Therefore,

$$\pi_A^* = (p_A^* - c)D_A^* = \left(t + \frac{\Delta q}{3}\right) \left(\frac{1}{2} + \frac{\Delta q}{6t}\right) = \frac{t}{2} + \frac{\Delta q}{3} + \frac{\Delta q^2}{18t}.$$

On the other hand,

$$\frac{(3t + \Delta q)^2}{18t} = \frac{9t^2 + 6t\Delta q + \Delta q^2}{18t} = \frac{t}{2} + \frac{\Delta q}{3} + \frac{\Delta q^2}{18t}$$

Hence $\pi_A^* = (3t + \Delta q)^2/(18t)$. The expression for π_B^* follows analogously. $\square$

Remark 1 (Relation to Bloch and Manceau 1999). Proposition 1 and Corollary 1 reproduce the standard Hotelling–Bertrand pricing result that is shared with Bloch and Manceau (1999), who analyze persuasive advertising by *private* firms in a differentiated-product setting. The key structural difference from their framework is that (i) the demand-shifting investment is undertaken by *public* actors (local government jurisdictions) rather than firms, (ii) spillovers ρ across jurisdictions are incorporated, (iii) welfare evaluation uses a welfare-relevant share α that discounts pure persuasion, and (iv) the first best is implemented via a single matching instrument, closing the loop from equilibrium distortion to intergovernmental-grant design. The downstream pricing game itself—and hence the reduced-form profit function—is identical.

4 Decentralized R&D

4.1 Preliminaries: expressing the perceived-quality differential

From (4),

$$q_A = g(e_1) + \rho g(e_2), \quad q_B = g(e_2) + \rho g(e_1),$$

so that

$$\Delta q \equiv q_A - q_B = (1 - \rho)(g(e_1) - g(e_2)). \quad (14)$$

4.2 Characterization of the symmetric Nash equilibrium

Assumption 1 (Uniqueness of best responses (sufficient concavity)). *On the interior region in Lemma 2, each jurisdiction's objective W_i^{loc} is strictly concave in its own choice e_i (so that best responses are unique).*¹¹

Proposition 2 (Symmetric Nash equilibrium and the decentralized FOC (with matching rate s)). *Under Assumption 1 and $s \leq 1$, the R&D game admits a symmetric equilibrium $e_1 = e_2 = e^N(s)$. If K' is increasing and g' is decreasing, the equilibrium is unique, and $e^N(s)$ satisfies*

$$(1 - s)K'(e^N(s)) = \frac{1 - \rho}{3} g'(e^N(s)). \quad (15)$$

Moreover, if $K'(0) = 0$ and $g'(0) > 0$, then $e^N(s) > 0$.

Proof. See Appendix B. On the interior region, reduced-form profits depend on (e_1, e_2) only through the perceived-quality differential Δq . Differentiating $W_1^{\text{loc}} = \pi_A^*(\Delta q) - (1 - s)K(e_1)$ by the chain rule yields a marginal benefit term proportional to $\partial \pi_A^* / \partial \Delta q$ times $\partial \Delta q / \partial e_1 = (1 - \rho)g'(e_1)$. Evaluating at the symmetric profile $e_1 = e_2$ implies $\Delta q = 0$, which delivers the decentralized FOC (15). The full proof, including the intermediate derivatives and the positivity claim, is in Appendix B. □

5 Welfare and Policy

5.1 Social welfare (fixing the evaluation criterion)

Social welfare is defined by (7). In accordance with the maintained evaluation criterion, welfare counts only the real component $r_f = \alpha q_f$ and excludes the persuasive component $a_f = (1 - \alpha)q_f$ (persuasion does not directly enter welfare).

We interpret α as a reduced-form welfare weight on perceived quality: $\alpha = 1$ corresponds to fully welfare-relevant (informative/real) quality, $\alpha = 0$ corresponds to purely persuasive demand shifting, and intermediate values allow partial welfare relevance.

¹¹A convenient sufficient condition for strict concavity (and hence uniqueness of best responses) is provided in Appendix B.

5.2 Preliminaries: market allocation and transport costs (interior region)

On the interior region, the indifferent location and the resulting market shares satisfy, by Proposition 1 and Lemma 1,

$$x^* = \frac{1}{2} + \frac{\Delta q}{6t}, \quad D_A^* = x^*, \quad D_B^* = 1 - x^*, \quad (\Delta q \equiv q_A - q_B). \quad (16)$$

The transport-cost term is

$$\int_0^1 t d_{\sigma(x)}(x) dx = t \int_0^{x^*} x dx + t \int_{x^*}^1 (1 - x) dx = \frac{t}{2}((x^*)^2 + (1 - x^*)^2).$$

Using the identity

$$(x^*)^2 + (1 - x^*)^2 = 2 \left(x^* - \frac{1}{2} \right)^2 + \frac{1}{2},$$

we can rewrite the transport-cost term as

$$\int_0^1 t d_{\sigma(x)}(x) dx = \frac{t}{4} + t \left(x^* - \frac{1}{2} \right)^2.$$

Finally, since (16) implies $x^* - 1/2 = \Delta q/(6t)$, we obtain

$$\int_0^1 t d_{\sigma(x)}(x) dx = \frac{t}{4} + \frac{(\Delta q)^2}{36t}. \quad (17)$$

5.3 The symmetric social optimum (constrained first best)

Terminological note. Because the planner takes the downstream market structure (Hotelling–Bertrand) and the technology g as given, the resulting optimum is a *constrained first best*: it is first-best conditional on the fixed market structure, not an unconstrained allocation.

Lemma 3 (Justifying symmetry of the social optimum). *Suppose jurisdictions 1 and 2 are symmetric and share the same primitives g and K . Assume the feasible set $[0, \bar{e}]^2$ is convex and the planner’s objective in (7) is concave in (e_1, e_2) on this set (so an optimum exists). Then the planner’s problem has at least one symmetric optimal solution with $e_1 = e_2$.*

Proof See Appendix C (Proof of Lemma 3). □

Proposition 3 (FOC for the symmetric social optimum (for the baseline model with α)). *By Lemma 3, it suffices to focus on symmetric policies $e_1 = e_2 = e$. If an interior*

solution exists and K' is increasing while g' is decreasing, then the symmetric social optimum $e^S(\alpha)$ is characterized by

$$K'(e^S(\alpha)) = \frac{\alpha(1+\rho)}{2} g'(e^S(\alpha)). \quad (18)$$

In particular, if $\alpha = 0$, then $e^S(0) = 0$.

Remark. When $\alpha = 0$ (purely persuasive quality), s^* in (19) is not defined; implementing $e^S(0) = 0$ generally requires a sufficiently negative matching rate (i.e., a tax) or an additional instrument, so we restrict attention to $\alpha > 0$ in Proposition 5. Note that $s^* \rightarrow -\infty$ as $\alpha \downarrow 0$, so we restrict attention to $\alpha > 0$ in Proposition 5.

Proof Under symmetry, $\Delta q = 0$ and the allocation is $D_A^* = D_B^* = 1/2$. Hence $q_A = q_B = (1+\rho)g(e)$ and welfare-relevant quality equals $\int_0^1 r_{\sigma(x)} dx = \alpha(1+\rho)g(e)$, while transport costs reduce to $t/4$ by (17). Dropping constants, $W^{\text{soc}}(e, e) = \alpha(1+\rho)g(e) - 2K(e) + \text{const}$, which yields (18). A complete derivation is provided in C. $\square$

5.4 A closed-form threshold for over- versus underinvestment

Proposition 4 (Real-quality share and the over/underinvestment threshold). *Compare the social optimum $e^S(\alpha)$ in Proposition 3 with the decentralized equilibrium $e^N(s)$ in Proposition 2. Let $\bar{\alpha}$ be defined in (1). When $s = 0$ (no matching), the following statements hold:*

$$\begin{aligned} \alpha < \bar{\alpha}(\rho) &\Rightarrow e^N(0) > e^S(\alpha), \\ \alpha = \bar{\alpha}(\rho) &\Rightarrow e^N(0) = e^S(\alpha), \\ \alpha > \bar{\alpha}(\rho) &\Rightarrow e^N(0) < e^S(\alpha). \end{aligned}$$

Thus, decentralization can entail overinvestment when α is small and underinvestment when α is large.¹²

Interpretation (business stealing vs. spillovers). Proposition 4 isolates a simple tug-of-war between two forces. On the one hand, when perceived quality is largely *persuasive* (small α), local governments spend to steal business from the neighbor, which shows up as a private incentive to raise e even though the welfare-relevant component is small—hence the tendency toward *overinvestment*. On the other hand, when perceived quality is largely *real* (large α), quality improvements generate spillovers across jurisdictions (through ρ) that are not fully internalized, so decentralization tends to *underinvest*. Figure 1 below turns this logic into a diagnostic chart.

Numerical illustration. Table 1 evaluates the baseline model under the log-quadratic specification $g(e) = \log(1+e)$, $K(e) = e^2/2$, $t = 1$. Several patterns emerge. When α is small relative to $\bar{\alpha}(\rho)$ (first two rows), decentralization overinvests— $e^N/e^S >$

¹²The specific cutoff $\bar{\alpha}(\rho) = 2(1-\rho)/[3(1+\rho)]$ depends on the Hotelling–Bertrand downstream structure and the linear spillover specification. Under alternative differentiation models (e.g., Salop circular city, logit demand, or Cournot), the sign-reversal structure—overinvestment for small α , underinvestment for large α —is expected to survive because it arises from the tension between business-stealing and spillover externalities, which is generic. However, the numerical value of the cutoff and the curvature of the boundary in (α, ρ) -space will differ; in this sense, the quantitative predictions are model-specific.

Table 1 Numerical examples: decentralized investment, social optimum, and optimal matching rate. Functional forms: $g(e) = \log(1 + e)$, $K(e) = e^2/2$, $t = 1$. The threshold for overinvestment is $\bar{\alpha}(\rho) = 2(1 - \rho)/[3(1 + \rho)]$.

α	ρ	$\bar{\alpha}(\rho)$	e^N	e^S	e^N/e^S	s^*	Regime
0.2	0.1	0.545	0.2416	0.1000	2.42	-1.727	over
0.2	0.5	0.222	0.1455	0.1325	1.10	-0.111	over
0.5	0.3	0.359	0.1952	0.2583	0.76	+0.282	under
0.8	0.1	0.545	0.2416	0.3307	0.73	+0.318	under
0.8	0.5	0.222	0.1455	0.4220	0.34	+0.722	under
1.0	0.3	0.359	0.1952	0.4487	0.44	+0.641	under
1.0	0.7	0.118	0.0916	0.5488	0.17	+0.882	under

1—and the corrective matching rate is negative (a tax). As α rises (rows 3–7), the regime switches to underinvestment and the matching rate becomes a subsidy. Higher spillovers ρ compress the threshold $\bar{\alpha}(\rho)$ toward zero, making underinvestment the more likely outcome and requiring larger subsidies.

5.5 Optimal design by the central government (implementation)

Proposition 5 (Implementation via matching grants). *Suppose $\alpha > 0$. On the relevant interior region, suppose Propositions 2 and 3 apply, and assume that K' is increasing and g' is decreasing. Then the constrained first best $e^S(\alpha)$ can be implemented in the decentralized equilibrium by setting the matching rate (possibly negative) $s \in (-\infty, 1]$ to*

$$s^* = 1 - \frac{\bar{\alpha}(\rho)}{\alpha}, \quad (19)$$

where $\bar{\alpha}(\rho)$ is defined in (1).¹³¹⁴

Proof By Proposition 2, the decentralized equilibrium is characterized by

$$(1 - s)K'(e^N(s)) = \frac{1 - \rho}{3} g'(e^N(s)).$$

Dividing both sides by $(1 - s)$ gives

$$K'(e^N(s)) = \frac{1 - \rho}{3(1 - s)} g'(e^N(s)).$$

¹³The matching rate s^* is financed through lump-sum transfers, so the baseline implementation problem abstracts from any additional deadweight loss of public finance. If, more realistically, public funds carry a shadow cost $\lambda > 1$ (reflecting distortionary taxation), the corrective benchmark would generally call for a weaker intervention, because the center would trade off the benefit of correcting the R&D externality against the cost of raising revenue. We do not solve that richer second-best problem here; the present formula should therefore be read as the clean benchmark under lump-sum finance.

¹⁴Throughout, we assume that the central government can *commit* to s before jurisdictions choose e_i . If commitment is not feasible—for example, because the center cannot resist ex post renegotiation—the equilibrium matching rate may differ from s^* . In a repeated-game setting, commitment might be sustained by reputation or institutional rules (e.g., multi-year grant programs with preset rates). The analysis of time-consistent matching policies is left for future work.

By Proposition 3, the social optimum satisfies

$$K'(e^S(\alpha)) = \frac{\alpha(1+\rho)}{2} g'(e^S(\alpha)).$$

Thus, implementing $e^N(s) = e^S(\alpha)$ requires

$$\frac{1-\rho}{3(1-s)} = \frac{\alpha(1+\rho)}{2} \iff 1-s = \frac{2(1-\rho)}{3\alpha(1+\rho)} \iff s = 1 - \frac{\bar{\alpha}(\rho)}{\alpha}.$$

which is exactly (19). $\square$

Alternative instruments when s^ is strongly negative.* As $\alpha \downarrow 0$, $s^* \rightarrow -\infty$: implementing the constrained first best requires arbitrarily large taxes on R&D—a setting in which the matching-grant mechanism is economically unrealistic. Three complementary policy tools may be considered for this regime: (i) *expenditure ceilings*: a direct cap on promotional spending (or equivalently a requirement that a minimum fraction of the budget be allocated to welfare-relevant research); (ii) *ex ante review*: a competitive-grant system in which a central agency evaluates proposed projects and funds only those whose α exceeds a threshold; and (iii) *performance-based transfers*: matching payments conditioned on measurable outcomes (e.g., yields, nutritional content) rather than expenditure. All three approaches restrict or screen the *composition* of investment, complementing—or substituting for—the *level* correction delivered by a matching rate.

5.6 A two-input extension: microfounding α via real R&D and promotion

The baseline model and the present extension play different but complementary roles. In Sections 2–5.5, the policy problem is one-dimensional and α is treated as an exogenous reduced-form summary of how much of the demand shifter is welfare-relevant; within that benchmark, the single matching rule $s^* = 1 - \bar{\alpha}(\rho)/\alpha$ provides a closed-form implementation result. Section 5.6 moves to a richer environment in which the composition of policy is itself endogenous, so that α is no longer primitive; in that setting, the baseline formula should be read as a benchmark for the one-instrument problem, whereas full efficiency generally requires separate instruments for research and promotion.

We keep the same downstream environment and the reduced-form profit dependence on the perceived-quality gap Δq (Corollary 1). We now decompose perceived quality into a welfare-relevant research component and a welfare-irrelevant promotional component:

$$q_i = q_i^R + q_i^M, \quad q_i^R = r_i + \rho_R r_j, \quad q_i^M = m_i + \rho_M m_j, \quad i \neq j,$$

where $\rho_R, \rho_M \in [0, 1)$ capture spillovers in each component. Local governments choose (r_i, m_i) and incur quadratic costs. The decentralized objective is

$$W_i^G = \pi_i^*(\Delta q) - \frac{k_R}{2} r_i^2 - \frac{k_M}{2} m_i^2,$$

where $\pi_i^*(\Delta q)$ is the same reduced-form profit as in (13). Social welfare follows the maintained criterion: only the real component enters welfare directly (promotion is pure persuasion), while market allocation depends on $q_i = q_i^R + q_i^M$ through Δq .

Proposition 6 (Two-input allocations under decentralization and under the planner). *Consider the interior region $|\Delta q| < 3t$ and suppose each government's objective is strictly concave in (r_i, m_i) (Appendix D provides a sufficient condition). Then the symmetric decentralized equilibrium satisfies*

$$r^N = \frac{1 - \rho_R}{3k_R}, \quad m^N = \frac{1 - \rho_M}{3k_M}.$$

The symmetric social optimum satisfies

$$r^S = \frac{1 + \rho_R}{2k_R}, \quad m^S = 0.$$

Hence decentralization underprovides welfare-relevant research ($r^N < r^S$) and overprovides purely persuasive promotion ($m^N > m^S$).

Corollary 2 (Endogenous welfare-relevant share (promotion tilt)). *In the symmetric decentralized equilibrium, the implied welfare-relevant share of perceived quality is*

$$\alpha^N \equiv \frac{q^R}{q^R + q^M} = \frac{k_M(1 - \rho_R^2)}{k_M(1 - \rho_R^2) + k_R(1 - \rho_M^2)}.$$

In particular, a higher spillover in real research (ρ_R) reduces α^N (a stronger tilt toward persuasion), strengthening the promotion-policy trap interpretation of the baseline threshold analysis.

Corollary 3 (Comparative statics of the endogenous welfare-relevant share). *Under the same conditions as Corollary 2:*

$$\frac{\partial \alpha^N}{\partial k_R} < 0, \quad \frac{\partial \alpha^N}{\partial k_M} > 0, \quad \frac{\partial \alpha^N}{\partial \rho_R} < 0, \quad \frac{\partial \alpha^N}{\partial \rho_M} > 0.$$

Costlier research or larger research spillovers tilt the policy mix toward promotion (lower α^N), while costlier promotion or larger promotion spillovers tilt it toward research (higher α^N).

Proof Direct differentiation of α^N in Corollary 2. In particular,

$$\frac{\partial \alpha^N}{\partial \rho_R} = -\frac{2\rho_R k_M k_R (1 - \rho_M^2)}{[k_M(1 - \rho_R^2) + k_R(1 - \rho_M^2)]^2} < 0, \quad \frac{\partial \alpha^N}{\partial \rho_M} = \frac{2\rho_M k_M k_R (1 - \rho_R^2)}{[k_M(1 - \rho_R^2) + k_R(1 - \rho_M^2)]^2} > 0.$$

The signs for k_R and k_M are analogous. $\square$

Economic interpretation. The result $\partial\alpha^N/\partial k_M > 0$ suggests that any development—such as the growth of social media—that *lowers* the unit cost of promotional activity ($k_M \downarrow$) will deepen the promotion tilt and lower α^N , thereby pushing the economy further into the overinvestment (promotion-trap) region of Figure 1.

Proposition 7 (Two-instrument implementation in the two-input model). *Consider the two-input model of Proposition 6. Suppose a central government can set separate matching rates s_R and s_M on research and promotion costs, so that jurisdiction i faces effective costs $(1 - s_R)\frac{k_R}{2}r_i^2 + (1 - s_M)\frac{k_M}{2}m_i^2$. Then:*

(i) *The symmetric decentralized equilibrium with matching is*

$$r^N(s_R) = \frac{1 - \rho_R}{3k_R(1 - s_R)}, \quad m^N(s_M) = \frac{1 - \rho_M}{3k_M(1 - s_M)}.$$

(ii) *The socially optimal research level $r^S = (1 + \rho_R)/(2k_R)$ is implemented by setting*

$$s_R^* = 1 - \frac{2(1 - \rho_R)}{3(1 + \rho_R)} = 1 - \bar{\alpha}(\rho_R), \quad (20)$$

where $\bar{\alpha}(\cdot)$ is the baseline threshold defined in (1). This has the same structural form as the baseline matching rate s^* in (19) with $\alpha = 1$ (because research is 100% welfare-relevant) and $\rho = \rho_R$.

(iii) *No finite matching rate s_M implements $m^S = 0$. Social welfare is strictly decreasing in s_M (since any promotion is pure waste), so the constrained optimum requires either (a) a direct ban on publicly funded promotion ($m_i = 0$), or (b) a sufficiently negative s_M (a confiscatory tax on promotion costs) to drive $m^N(s_M)$ close to zero.*

Proof Part (i) follows from the FOC of jurisdiction 1 at symmetry:

$$\frac{1 - \rho_R}{3} = (1 - s_R)k_R r \quad \Rightarrow \quad r = \frac{1 - \rho_R}{3k_R(1 - s_R)},$$

and analogously for m . Part (ii) equates $r^N(s_R) = r^S$:

$$\frac{1 - \rho_R}{3k_R(1 - s_R)} = \frac{1 + \rho_R}{2k_R} \iff 1 - s_R = \frac{2(1 - \rho_R)}{3(1 + \rho_R)} = \bar{\alpha}(\rho_R).$$

Part (iii): Since $m^N(s_M) = (1 - \rho_M)/[3k_M(1 - s_M)] > 0$ for all finite s_M (as $\rho_M < 1$), no finite rate implements $m^S = 0$. Moreover, social welfare is $W^{\text{soc}} = (1 + \rho_R)r - k_R r^2 - k_M m^2 + \text{const}$, and $\partial W/\partial s_M = (\partial W/\partial m) \cdot (\partial m^N/\partial s_M) < 0$ because $\partial W/\partial m = -2k_M m < 0$ and $\partial m^N/\partial s_M > 0$. $\square$

Remark. The asymmetry between (ii) and (iii) is instructive: matching grants are well suited to correcting the *level* of a welfare-relevant input (research) but cannot eliminate a welfare-irrelevant input (promotion) whose private return is strictly positive. This makes matching grants a necessary but insufficient policy tool in the two-input setting; a quantity regulation or institutional separation (ring-fencing research budgets from promotional budgets) is needed to close the gap.

The extension clarifies the baseline role of α : it summarizes the composition of policy between welfare-relevant research and welfare-irrelevant promotion. Even when jurisdictions “invest a lot” in the demand shifter q , decentralization can be socially wasteful because it reallocates spending toward persuasion and lowers the welfare-relevant share.

6 Conclusion

This paper develops a deliberately minimal model of differentiation competition to study an institutional arrangement in which local governments supply product-oriented R&D (e.g., cultivar breeding and related branding investments) and downstream firms (cooperatives, trading firms, and jurisdiction-level marketing/branding bodies) engage in price competition using the resulting assets. The analysis delivers closed-form conditions under which decentralized public R&D can reduce social welfare, and it provides a transparent corrective benchmark. The central step is to recognize that while jurisdictional R&D raises consumers’ perceived utility through a demand shifter (perceived quality), that shift need not map one-for-one into real welfare. We therefore decompose perceived quality using the real-quality share α and then leverage the closed-form downstream price equilibrium and reduced-form profits to compare the decentralized R&D equilibrium with the social planner’s allocation. The resulting comparison yields a sharp threshold: when α lies below $\bar{\alpha} = \frac{2(1-\rho)}{3(1+\rho)}$, decentralization can generate excessive investment, whereas for α above this threshold decentralization can generate underinvestment.

The policy implication is that one should not evaluate interjurisdictional R&D competition uniformly. Instead, the appropriate degree and direction of coordination—support or restraint—depends on (i) whether the relevant R&D primarily creates persuasion (demand reallocation) or real quality (a genuine welfare improvement), captured by α , and (ii) the magnitude of knowledge spillovers, ρ .¹⁵ Put differently, the same decentralized “race” can be wasteful when it is mostly business stealing (low α) but insufficient when it is mostly true quality with external benefits (high α). Figure 1 provides a simple policy-diagnostic mapping: the region $\alpha < \bar{\alpha}(\rho)$ calls for disciplining instruments—a lower/possibly negative matching rate (which can be interpreted as restraint, including tax-like discipline)—whereas $\alpha > \bar{\alpha}(\rho)$ calls for support (a positive matching rate), and the illustrative cases discussed in the introduction can be placed on the chart depending on whether they are closer to persuasion-heavy branding or research-heavy quality improvement. In the baseline model with a single policy margin, a central government (or any effective coordination mechanism) can use a single matching instrument applied to R&D costs as a closed-form benchmark for corrective design, implementing the constrained social optimum within that one-instrument environment. In the two-input extension that separates real research

¹⁵Empirically, the dominant finding in the agricultural R&D literature is that public research earns high rates of return—median estimates range from 40% to over 60% per year (Hurley et al. 2014; Alston et al. 2020)—which suggests that underinvestment (high α , strong spillovers) is the more prevalent case for genuine research. By contrast, evidence on the returns to generic promotion is mixed and generally lower (Kinnucan and Zheng 2004). In terms of our model, this empirical landscape points to a world in which most jurisdictions sit in the *underinvestment* region ($\alpha > \bar{\alpha}$), and a positive matching rate is warranted—though the promotion component of public programs may still lie in the *overinvestment* region.

from purely persuasive promotion, decentralization can also distort the composition of policy toward persuasion; fully restoring efficiency generally requires instrument differentiation across policy components (e.g., supporting research while disciplining promotion). The analysis thus highlights the complementarity between (i) cooperative mechanisms such as the division of research tasks, joint research, and systematic sharing of outcomes, and (ii) an evaluation framework that disciplines whether public interventions are producing real quality improvements rather than merely reallocating demand through persuasive channels.

Two caveats are worth noting. First, our results are derived in a deliberately stylized two-jurisdiction Hotelling environment; we do not claim that the closed-form cutoff or the implementation formula are quantitatively robust to richer geography or additional jurisdictions. That said, the core mechanism is a general one: decentralized incentives are pushed toward overprovision by business-stealing motives when perceived quality is largely persuasive (low α), and toward underprovision when welfare-relevant quality generates spillovers (high ρ); extensions to three regions or circular-city settings are expected to preserve this basic trade-off even if the exact threshold shifts.

Second, implementing s^* is information intensive: it requires the center to assess the welfare-relevant share α and the spillover parameter ρ , which are likely hard to observe and may be mismeasured. A natural implication of the model is that misestimation can lead not only to the wrong intensity but even the wrong *direction* of intervention (supporting when restraint is warranted, or restraining when support is warranted). Finally, while the formula allows “negative matching,” in practice this can be interpreted less as a literal tax on local R&D and more as regulatory discipline—tight eligibility definitions, caps on subsidized spending, and disclosure/evaluation requirements that screen out persuasion-heavy programs.

Because the paper intentionally focuses on a stripped-down benchmark, several extensions are natural. First, rather than taking α as exogenous, one could model institutional choices—certification schemes, third-party evaluation, and disclosure requirements—that endogenously shift the relative weight of the real component, thereby clarifying the interaction between evaluation institutions and R&D competition. Second, introducing export markets or demand from outside the two jurisdictions would bring in welfare-relevant market-expansion effects and international differentiation competition, sharpening the role of coordination, intellectual-property and branding governance, and the policy consequences of spillovers. Third, endogenizing downstream organizational forms (integration of cooperatives, competition among trading firms, and governance of collective brands) would allow a more direct analysis of the interaction between the institutional design that mitigates cannibalization and the incentives generated by interjurisdictional R&D competition. These extensions can build directly on the closed-form building blocks derived here—the price equilibrium, reduced-form profits, decentralized and planner first-order conditions, and the implementation result—and add comparative statics and welfare evaluation in richer environments.

In sum, local-government R&D and brand formation are potentially powerful instruments for regional differentiation; yet when they primarily operate through persuasion (business stealing), decentralized promotion policies may induce excessive and socially wasteful investment. This paper characterizes the conditions under which the direction of the distortion reverses and provides a design principle that implements the social optimum using a minimal policy instrument.

A Proofs for Section 3

Proof of Lemma 2 (1) If $e_1 = e_2 = \bar{e}$, then (4) implies $q_A = g(\bar{e}) + \rho g(\bar{e}) = q_B$, hence $\Delta q = 0$. Substituting $\Delta q = 0$ into Proposition 1 yields $p_A^* = p_B^* = c + t$ and $D_A^* = D_B^* = 1/2$, which is interior.

(2) The mapping $\Delta q(e_1, e_2) = (1 - \rho)\{g(e_1) - g(e_2)\}$ is continuous because g is continuous, and it satisfies $\Delta q(\bar{e}, \bar{e}) = 0$. By the definition of continuity, there exists $\varepsilon > 0$ such that $\|(e_1, e_2) - (\bar{e}, \bar{e})\| < \varepsilon$ implies $|\Delta q(e_1, e_2)| < 3t$. This inequality guarantees interior demand (equivalently $D_A^* \in (0, 1)$), so Proposition 1 and Corollary 1 apply in this neighborhood.

(3) The bound follows immediately from $\Delta q = (1 - \rho)\{g(e_1) - g(e_2)\}$ and $e_i \in [0, \bar{e}]$. If (9) holds, then $|\Delta q| < 3t$ for every feasible profile, which in turn justifies uniqueness of the interior equilibrium and the global use of the reduced-form profits and chain-rule differentiation. $\square$

Proof of Corollary 1 Let $\Delta q \equiv q_A - q_B$. By Proposition 1,

$$p_A^* - c = t + \frac{\Delta q}{3}, \quad D_A^* = \frac{1}{2} + \frac{\Delta q}{6t}.$$

Therefore,

$$\pi_A^* = (p_A^* - c)D_A^* = \left(t + \frac{\Delta q}{3}\right) \left(\frac{1}{2} + \frac{\Delta q}{6t}\right) = \frac{t}{2} + \frac{\Delta q}{3} + \frac{\Delta q^2}{18t}.$$

On the other hand,

$$\frac{(3t + \Delta q)^2}{18t} = \frac{9t^2 + 6t\Delta q + \Delta q^2}{18t} = \frac{t}{2} + \frac{\Delta q}{3} + \frac{\Delta q^2}{18t}.$$

Hence $\pi_A^* = (3t + \Delta q)^2/(18t)$. The expression for π_B^* follows analogously. $\square$

Proof of Proposition 1 By Lemma 1, on the interior region demands are $D_A = x^*$ and $D_B = 1 - x^*$. Hence profits are

$$\begin{aligned} \pi_A &= (p_A - c) \frac{q_A - q_B - p_A + p_B + t}{2t}, \\ \pi_B &= (p_B - c) \frac{-q_A + q_B + p_A - p_B + t}{2t}. \end{aligned}$$

Consider firm A. Since $\partial D_A / \partial p_A = -1/(2t)$, its first-order condition is

$$\frac{\partial \pi_A}{\partial p_A} = D_A + (p_A - c) \frac{\partial D_A}{\partial p_A} = \frac{q_A - q_B + c - 2p_A + p_B + t}{2t}.$$

Analogously,

$$\frac{\partial \pi_B}{\partial p_B} = \frac{-q_A + q_B + c + p_A - 2p_B + t}{2t}.$$

The second-order conditions hold because

$$\frac{\partial^2 \pi_A}{\partial p_A^2} = -\frac{1}{t} < 0, \quad \frac{\partial^2 \pi_B}{\partial p_B^2} = -\frac{1}{t} < 0.$$

Solving the first-order conditions,

$$q_A - q_B + c - 2p_A + p_B + t = 0, \quad -q_A + q_B + c + p_A - 2p_B + t = 0,$$

yields the unique Nash equilibrium,

$$p_A^* = c + t + \frac{q_A - q_B}{3}, \quad p_B^* = c + t - \frac{q_A - q_B}{3}.$$

Substituting into (8) delivers (12). $\square$

B Proofs for Section 4

A sufficient condition for strict concavity

On the interior region, π_i^* is a quadratic function of Δq as in (13), with $\Delta q = (1 - \rho)\{g(e_i) - g(e_j)\}$ by (14). Holding e_j fixed and differentiating yields

$$\frac{\partial^2 \pi_i^*}{\partial e_i^2} = \frac{1}{9t} \left((1 - \rho)^2 \{g'(e_i)\}^2 + (1 - \rho) \{3t \pm \Delta q\} g''(e_i) \right) \leq \frac{(1 - \rho)^2}{9t} \{g'(e_i)\}^2,$$

where the inequality uses $g'' \leq 0$. Thus, under $s < 1$, if

$$(1 - s)K''(e_i) > \frac{(1 - \rho)^2}{9t} \{g'(e_i)\}^2 \quad \text{for all } e_i \in [0, \bar{e}],$$

then $W_i^{\text{loc}} = \pi_i^* - (1 - s)K$ is strictly concave in e_i . For instance, if $g(e) = \beta \log(1 + e)$ and $K(e) = \frac{\kappa}{2}e^2$, then $g'(e) \leq \beta$ implies that $(1 - s)\kappa > \frac{(1 - \rho)^2}{9t}\beta^2$ is sufficient.

Proof of Proposition 2 By Lemma 2, on the interior region the reduced-form profits in Corollary 1 apply, and differentiation via the chain rule is valid. Jurisdiction 1 maximizes

$$W_1^{\text{loc}}(e_1, e_2) = \pi_A^*(q_A, q_B) - (1 - s)K(e_1).$$

Differentiating with respect to e_1 gives

$$\frac{\partial W_1^{\text{loc}}}{\partial e_1} = \frac{\partial \pi_A^*}{\partial \Delta q} \cdot \frac{\partial \Delta q}{\partial e_1} - (1 - s)K'(e_1). \quad (21)$$

From Corollary 1,

$$\pi_A^* = \frac{(3t + \Delta q)^2}{18t} \Rightarrow \frac{\partial \pi_A^*}{\partial \Delta q} = \frac{3t + \Delta q}{9t}.$$

Moreover, (14) implies

$$\frac{\partial \Delta q}{\partial e_1} = (1 - \rho)g'(e_1).$$

Substituting into (21) yields

$$\frac{\partial W_1^{\text{loc}}}{\partial e_1} = \frac{3t + \Delta q}{9t} (1 - \rho)g'(e_1) - (1 - s)K'(e_1).$$

At a symmetric profile $e_1 = e_2 = e$, we have $\Delta q = 0$, so

$$\left. \frac{\partial W_1^{\text{loc}}}{\partial e_1} \right|_{e_1=e_2=e} = \frac{3t}{9t} (1 - \rho)g'(e) - (1 - s)K'(e) = \frac{1 - \rho}{3} g'(e) - (1 - s)K'(e).$$

Imposing the first-order condition $\partial W_1^{\text{loc}} / \partial e_1 = 0$ gives (15). If $K'(0) = 0$ and $g'(0) > 0$, then the right-hand side of (15) is strictly positive at $e = 0$ whereas the left-hand side equals zero; monotonicity therefore implies $e^N(s) > 0$. Uniqueness follows from Assumption 1 together with standard monotonicity arguments. $\square$

C Proofs for Section 5

Proof of Lemma 3 The welfare function (7) is invariant to relabeling ($1 \leftrightarrow 2$). Hence, if (e_1, e_2) is optimal, then so is the swapped profile (e_2, e_1) , yielding the same welfare level. If the feasible set is convex and the objective is concave in (e_1, e_2) (a convenient sufficient condition), then the average profile $(\frac{e_1+e_2}{2}, \frac{e_1+e_2}{2})$ achieves at least the same welfare. Therefore, a symmetric optimum exists. $\square$

Proof of Proposition 3 Under a symmetric policy $e_1 = e_2 = e$, (4) implies

$$q_A = q_B = (1 + \rho)g(e), \quad \Delta q = 0, \quad D_A^* = D_B^* = \frac{1}{2}.$$

Moreover, by (3), $r_A = r_B = \alpha(1 + \rho)g(e)$, so

$$\int_0^1 r_{\sigma(x)} dx = r_A D_A^* + r_B D_B^* = \alpha(1 + \rho)g(e) \cdot \frac{1}{2} + \alpha(1 + \rho)g(e) \cdot \frac{1}{2} = \alpha(1 + \rho)g(e).$$

Transport costs equal $t/4$ by substituting $\Delta q = 0$ into (17). Dropping constants, we can write

$$W^{\text{soc}}(e, e) = \alpha(1 + \rho)g(e) - 2K(e) - \frac{t}{4} + \text{const.}$$

The first-order condition is therefore

$$\frac{d}{de} W^{\text{soc}}(e, e) = \alpha(1 + \rho)g'(e) - 2K'(e) = 0,$$

which is equivalent to (18). When $\alpha = 0$, the condition reduces to $-2K'(e) = 0$, and since $K'(e) > 0$ for $e > 0$, it follows that $e^S(0) = 0$. $\square$

Appendix D Proofs for Section 5.6 (two-input extension)

D.1 Reduced-form profit derivative

From Corollary 1, for seller A the reduced-form equilibrium profit on the interior region is $\pi_A^*(\Delta q) = (3t + \Delta q)^2 / (18t)$, hence

$$\frac{\partial \pi_A^*}{\partial \Delta q} = \frac{3t + \Delta q}{9t}, \quad \left. \frac{\partial \pi_A^*}{\partial \Delta q} \right|_{\Delta q=0} = \frac{1}{3}.$$

D.2 Perceived-quality gap in (r, m)

By construction,

$$\Delta q = (q_i^R + q_i^M) - (q_j^R + q_j^M) = (1 - \rho_R)(r_i - r_j) + (1 - \rho_M)(m_i - m_j),$$

so $\partial \Delta q / \partial r_i = 1 - \rho_R$ and $\partial \Delta q / \partial m_i = 1 - \rho_M$.

D.3 Decentralized FOCs and symmetric equilibrium

Government i maximizes

$$W_i^G = \pi_i^*(\Delta q) - \frac{k_R}{2}r_i^2 - \frac{k_M}{2}m_i^2.$$

Using the chain rule, evaluated at the symmetric point $\Delta q = 0$,

$$0 = \frac{\partial W_i^G}{\partial r_i} = \frac{\partial \pi_i^*}{\partial \Delta q} \Big|_{\Delta q=0} (1 - \rho_R) - k_R r_i \Rightarrow r^N = \frac{1 - \rho_R}{3k_R},$$

and analogously $m^N = (1 - \rho_M)/(3k_M)$.

D.4 Concavity condition (sufficient)

Since $\pi_i^*(\Delta q)$ is convex in Δq with $\partial^2 \pi_i^*/\partial \Delta q^2 = 1/(9t)$ and Δq is linear in (r_i, m_i) , the Hessian of W_i^G with respect to (r_i, m_i) is

$$H_i = \frac{1}{9t} \begin{pmatrix} (1 - \rho_R)^2 & (1 - \rho_R)(1 - \rho_M) \\ (1 - \rho_R)(1 - \rho_M) & (1 - \rho_M)^2 \end{pmatrix} - \begin{pmatrix} k_R & 0 \\ 0 & k_M \end{pmatrix}.$$

A sufficient (and in this rank-one form, also necessary) condition for strict concavity is

$$\frac{1}{9t} \left(\frac{(1 - \rho_R)^2}{k_R} + \frac{(1 - \rho_M)^2}{k_M} \right) < 1.$$

D.5 Social optimum and $m^S = 0$

Under the maintained welfare criterion, promotion does not enter welfare directly. At the symmetric point, $\Delta q = 0$, so the allocation and transport-cost terms are locally unaffected to first order by a symmetric change in m . Therefore,

$$\frac{\partial W^S}{\partial m_i} \Big|_{\text{sym}} = -k_M m_i \Rightarrow m^S = 0.$$

For research, at symmetry each jurisdiction serves half of consumers and a marginal increase in r_i raises the real component for its own consumers by 1 and for the other jurisdiction's consumers via the spillover by ρ_R . Hence the marginal social gain is $(1 + \rho_R)/2$, yielding

$$\frac{\partial W^S}{\partial r_i} \Big|_{\text{sym}} = \frac{1 + \rho_R}{2} - k_R r_i \Rightarrow r^S = \frac{1 + \rho_R}{2k_R}.$$

D.6 Endogenous share

At symmetry, $q^R = (1 + \rho_R)r^N$ and $q^M = (1 + \rho_M)m^N$, hence

$$\alpha^N = \frac{(1 + \rho_R)r^N}{(1 + \rho_R)r^N + (1 + \rho_M)m^N} = \frac{k_M(1 - \rho_R^2)}{k_M(1 - \rho_R^2) + k_R(1 - \rho_M^2)}.$$

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Author Contributions

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